Course Outline

1. Course Identity

A. Course as listed in CUHK-Shenzhen

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	MDS5122
Course title (English)	Deep Learning and Their Applications
Course title (Chinese)	深度学习及其应用
Units	3
Language of Instruction	English
Description (English)	This course will cover key new techniques intended to help students and practitioners enter the field of machine learning in general and deep learning in particular. The topics include multilayer perceptron (MLP), stacked autoencoders (denoising, variational and adversarial), convolutional neural networks (ResNet, Inception)，recurrent neural networks (LSTM, GRU), multi-task deep learning, transfer learning, deep generative models and deep Gaussian process. Selected use cases include sentiment analysis，chatbot，image based document classification and machine translation.
Description (Chinese)	深度学习与应用是旨在从实践的角度阐述现代深度神经网络的基本组成部分以及相关的应用领域。课程涵盖的主题包括卷积神经网络、循环神经网络、图神经网络、神经辐射场、注意力机制和 Transformer。 在此基础上，本课程还引入了一些精心挑选的计算机视觉和 既然语言处理 应用任务进行介绍。 课程实施的过程中将安排一系列实践项目来巩固和强化课程中学到的概念。 通过学习本课程，学生可以掌握使用python库构建深度神经网络并解决一些实际中常见的人工智能任务。

B. Corresponding course in CUHK

Please give details of the *closest* corresponding course in CUHK (as approved by CUHK Senate and listed in course list). If the course in Shenzhen maps to more than one course in CUHK, please make multiple copies of the block below.

Course code	ELEG5491
Course title (English)	Introduction to Deep Learning
Course title (Chinese)	
Units	3
Description (English)	This course provides an introduction to deep learning. Students

	taking this course will learn the theories, models, algorithms, implementation and recent progress of deep learning, and obtain empirical experience on training deep neural networks. The course starts with machine learning basics and some classical deep models (including convolutional neural network, deep belief net, and auto-encoder), followed by optimization techniques for training deep neural networks, implementation of large-scale deep learning, multi-task deep learning, transferred deep learning, recurrent neural networks, applications of deep learning to computer vision and speech recognition, and understanding why deep learning works. The students taking are expected to have some basic background knowledge on calculus, linear algebra, probability, statistics and random process as a prerequisite.
Description (Chinese)	

2. Prerequisites / Co-requisites

Please state prerequisites and co-requisites, in terms of courses in CUHK-Shenzhen* or any other requirements (e.g., having taken certain subjects in high school).

(* Because course codes may not yet be stable, please provide both course code and course title.)

A. Prerequisites

CSC1001 or CSC1003, DDA3020/CSC4020/DDA2020/FTE4560
Remark: Requiring DDA4250 knowledge and solid programming skills

B. Co-requisites

3. Learning Outcomes

1. Understand the fundamental concepts, theories and algorithms of deep learning. (K/S);
2. Understand the principles of designing different deep models for various applications (K/S);
3. Implement and train deep neural networks with deep learning toolboxes (K/S);
4. Obtain empirical experience of designing and training deep models in practical applications. (K/S/V).

4. Course syllabus

1.Introduction
Giving the brief go through ofthe basic plan and goal ofthis course; Summarizing the different topics we will cover in the course; Introducing the assignments/projects ofthis course (the students will select to complete); Introducing the history of AI and the machine learning basics.

2. Convolutional Neural Networks
Introducing Convolutional Layer, Pooling Layer, Normalization Layer (from whitening to BN, IN, LN); AlexNet, VGG, InceptionNet; Skip Connection, ResNet and DenseNet; Giving the concept of receptive field; Optimization-2 (Momentum, Adam); Illustrating how to build up a machine learning model by using Pytorch.

3. Computer Vision Basics
Introducing the concept of multi-scale modeling; Image Classification, Object Detection, Segmentation, 3D-Conv and Action Recognition; Introducing the OpenMMLab framework for building up compute vision system.

4. Recurrent Neural Networks
Introducing some basic RNN architecture, including GRU, Recurrent Neural Network, LSTM, BiRNN, as well as Recursive Neural Network.

5. NLP basics
Introducing some basic concepts of NLP, such as Word Embedding, word2vec, Glove, seq2seq model, and some standard NLP tasks, such as Text Sentiment Classification and Machine Translation; Introducing the NLTK framework for building up NLP system.

6. Graph Neural Network
Introducing some basic concepts about GNN, including Graph Embedding, DeepWalk, Node2vec, Struc2vec, SDNE, GCN and GraphSAGE (graph sample and aggregation); Introducing the PyG library for GNN practice.

7. Attention and Transformer
Introducing some basic concepts of Attention Mechanism and Transformer. Discussing their application in NLP, including BERT, Masked Language Modeling; Discussing their application in CV, including Non-local in CNN, ViT architecture, Swin Transformer; Autoencoder, and Masked Autoencoder.

8. Self-supervised Learning
Introducing some basic concepts about self-supervised training, including Unsupervised and selfsupervised learning, Contrastive Learning, Generative methods, Dual Learning, etc.

9. Unify vision and language
Introducing Cross-modal Transformer (network perspective), Contrastive Language-

Image Pretraining (training perspective), Image Captioning, Visual Grounding, and Text2Image.

10. Neural Radiance Fields
Introduction to the basic technique ofNeural Radiance Fields, such as Positional encoding, The radiance field function approximator, Differentiable volume renderer, Stratified sampling, Hierarchical volume sampling.

5. Assessment Scheme

Component/ method	**% weight**
Assignment 1	20
Assignment 2	20
Final Exam	20
Final Project	40

6. Grade descriptor

General

Grade	**Description**
A/A-	Eceptionally good performance and far exceeding expectation in all or most of the course learning outcomes; Demonstration of superior understanding ofthe subject matter, the ability to analyze problems and apply extensive knowledge, and skillful use of concepts and materials to derive proper solutions.
B	Good performance in all course learning outcomes and exceeding expectation in some ofthem; Demonstration of a good understanding ofthe subject matter and the ability to use proper concepts and python tools to solve most of the problems encountered in a practical way.
C	Adequate performance and meeting expectations in all course learning outcomes; Demonstration of adequate understanding of the subject matter and the ability 4 to solve simple tasks.
D	Performance barely meets the expectation in the essential course learning outcomes; Demonstration of partial understanding of the subject matter and the ability to solve simple tasks.
F	Demonstration of a poor understanding ofthe course materials. Non-ability in applying some basic tools to solve simple problems.

7. Feedback for evaluation

Formal Course and Teaching Evaluation
Informal feedback to instructor and/or teaching assistant

8. Reading

Recommended Reading:

9. Course components

Activity	Hours/week
Lecture	3
Tutorial	1

10. Indicative teaching plan

Week	Content/ topic/ activity
1	Course Introduction / Giving the brief go through of the basic plan and goal of this course; Summarizing the different topics we will cover in the course; Course Introduction / Giving the brief go through of the basic plan and goal of this course; Summarizing the different topics we will cover in the course; Introducing the assignments/projects of this course; The machine learning basics.
2	Convolutional Neural Network 1 / Convolutional Layer, Pooling Layer, Normalization Layer (from whitening to BN, IN, LN); AlexNet, VGG, InceptionNet; Skip Connection, ResNet and DenseNet.
3	Convolutional Neural Network 2 / The concept of receptive field; MobileNet; ShuffleNet; Optimization-2 (Momentum, Adam); The Pytorch basic (how to build up a machine learning model by using Pytorch) .
4	Computer Vision Basics / The concept of multi-scale modeling; Image Classification, Object Detection, Segmentation, 3D-Conv and Action Recognition; Introducing the OpenMMLab framework for building up compute vision system.
5	Recurrent Neural Network / GRU; Recurrent Neural Network; LSTM; Bi- RNN; Recursive Neural Network.
6	NLP basics / Word Embedding, word2vec; Glove; seq2seq; Text Sentiment Classification; Machine Translation; Introducing the NLTK framework for building up NLP system
7	Graph Neural Network / Graph Embedding, DeepWalk, Node2vec, Struc2vec, SDNE;GCN; GraphSAGE (graph sample and aggregation); Introducing the PyG library for GNN practice
8	Attention and Transformer 1 / Attention Mechanism; Transformer in NLP; BERT; Masked Language Modeling (MLM)

9	Attention and Transformer 2 / Non-local in CNN; ViT architecture, Swin Transformer; Autoencoder; Masked Autoencoder
10	Self-supervised Learning / Unsupervised and self-supervised learning; Contrastive Learning; Generative methods; Dual Learning
11	Unify vision and language / Cross-modal Transformer (network perspective); Contrastive Language-Image Pre-training (training perspective); Image Captioning; Visual Grounding; Text2Image
12	Neural Radiance Fields / Positional encoding, The radiance field function approximator, Differentiable volume renderer, Stratified sampling, Hierarchical volume sampling.
13	Course Project and Group Representation 1
14	Group Representation 2 and course review

11. Implementation plan

The implementation plan may vary from year to year. Please indicate expected enrolment, and number of sections.

[Example: 150 students for lecture (x 2); 30 students for tutorials (x 10)]

150 students for lectures and tutorials

12. Any other information

None

13. Approval

Has the course title been included in the programme submission approved by CUHK Senate? Are there any differences?

Have the details (as in this document) been approved at School or other level in CUHK-Shenzhen?

Submit application

14. Version date

Version number	2
As of (date)	2025-08-05